Mohan Grewal

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EXPERIENCE

Software Engineer Intern | Android Studio, Xcode, Kotlin, Swift, Linux

Rivian — **January 2023 - August 2023, Vancouver, BC**

- Diagnosed and resolved customer-facing mobile app issues in Kotlin and Swift, **improving QA turnaround for production builds by 35%.**
- Engineered an in-app debug menu used by a QA team of 20+ to auto-generate **100+ structured JIRA tickets** with runtime device state and logs, eliminating manual bug report workflows.
- Refactored shared mobile modules to reduce duplicated logic across feature branches, **improving maintainability and decreasing regression defects** during release cycles.
- Developed an automated testing suite adapted from Python into Kotlin and integrated into developer workflow, **reducing manual feature validation by 25%** and improving test coverage.
- Instrumented structured logging and runtime diagnostics **to accelerate root-cause analysis** across cross-functional teams.
- Developed a CI-integrated pipeline to onboard new automated test cases into a **24-hour continuous QA testing suite**, improving nightly build validation and expanding automated regression coverage.
- Designed mobile telemetry hooks to capture real-time client-side failure states, **enabling reproducible issue tracking in staging and production environments.**

Software Engineer Intern | Unity C#

East Side Games — **September 2020 - April 2021, Vancouver, BC**

- Designed and shipped a successful weekly subscription service using Unity IAPs for both Android and iOS platforms, **deployed in production for a live mobile game with 5.5M+ installs.**
- Architected client-side purchase validation and subscription state management logic supporting **1000+ concurrent active subscriptions** across platforms.
- Implemented **structured logging and robust edge-case handling across purchase, renewal, and restore flows** to ensure subscription state consistency.
- Developed a visual debugging menu **actively used and iterated on 30+ internal QA and live-ops team members** to diagnose purchase failures and restore service issues in real time.
- Built safeguards for platform-specific billing edge cases (refunds, cancellations, renewals), **improving transactional reliability across iOS and Android storefronts by 20%**.
- Refactored shared client-side logic **into reusable modular components adopted across multiple runtime systems**, reducing duplicated state validation paths in production builds.
- Investigated and resolved production defects **by tracing asynchronous client-side service interactions through platform APIs**, improving reliability in shipped mobile builds.

PROJECT

Spotify Smart Queue Manager | Electron, React, TypeScript, JavaScript, Node.js, Spotify Web API

Personal · github.com/mohangrewal101/Spotify-Smart-Queue-Manager · August 2025 - January 2026

- Developed an Electron desktop widget **enabling real-time Spotify queue management** without interrupting playback or switching applications.
- Integrated the Spotify Web API and implemented custom logic **supporting multiple queue operations**, including reordering and removal.
- Optimized frontend rendering and backend state updates to ensure real-time queue reflection with **less than 150ms latency**, allowing Spotify track changes to appear instantly under concurrent operations.
- Designed IPC-based state synchronization between Node.js backend services and the React frontend to **prevent queue state desynchronization.**
- Implemented robust error handling and retry mechanisms for Spotify Web API requests, **improving queue reliability and reducing failure-induced desynchronizations** during live usage.

Panic Titanic | Unity, C#, Photon Networking

Personal · github.com/2020-21-UBC-Game-Dev-Team-3/team3-game · September 2020 - March 2021

- **Led development** of a real-time 3D multiplayer game using Unity and Photon Networking, coordinating a 10-person team in a sprint-based agile workflow and **winning the Audience Choice Award** at the UBC Game Development Competition.
- Implemented RPC-based state synchronization **to maintain consistent gameplay across networked clients** in real-time multiplayer sessions.
- Structured multiplayer game events using Photon's event-driven networking model **to ensure deterministic state updates across distributed clients.**
- Reduced network update frequency and payload size to improve gameplay responsiveness across **10 concurrent players** per multiplayer session.
- Designed and integrated a multiplayer voting system **using C# enums and Photon callbacks** to coordinate real-time player decisions across distributed clients, **praised for clear and reliable player decision coordination.**

AI Workout Plan | FastAPI, Python, React, JavaScript, Tailwind CSS, scikit-learn

Personal · github.com/mohangrewal101/Workout-AI-Planner · April 2025 - June 2025

- Built and deployed a full-stack AI-powered workout planner that **generates personalized routines and recommendations** based on user-selected goals and muscle groups.
- Constructed and labeled a custom training dataset by mapping exercises and fitness tips to target muscle groups, movement patterns, and user goals **to enable supervised recommendation modeling.**
- **Trained a logistic regression model** using scikit-learn to rank and filter **40+ fitness tips**, improving recommendation relevance by muscle group focus.
- Designed RESTful APIs using FastAPI to **efficiently orchestrate frontend requests, ML inference, and data retrieval,** enabling low-latency personalized recommendations.
- Implemented backend feature transformation logic to convert user-selected inputs into structured model-ready representations, **enabling consistent real-time recommendation inference.**

Path-Reachability | IntelliJ, TypeScript, Javascript, Z3

Academic · github.com/mohangrewal101/Path-Reachability · March 2023 - April 2024

- Developed a **static analysis tool** to enumerate **all feasible execution paths** in TypeScript programs, enabling early detection of unreachable and conflicting logic.
- Designed parsed execution paths into a structured AST representation to **making program behavior interpretable and usable by downstream tooling.**
- Generated **solver-ready path constraints from AST traversal** to validate execution paths in TypeScript programs.
- Modeled and evaluated execution path constraints using the Z3 solver, **classifying 100+ path conditions as satisfiable or unsatisfiable** to simplify visualization and analysis in the UI layer.
- Implemented **constraint pruning to eliminate infeasible execution paths prior to solver evaluation**, reducing unnecessary satisfiability checks during analysis.

EDUCATION

Software Systems Developer Certificate

British Columbia Institute of Technology · Vancouver, BC · 2026

Bachelor of Science in Computer Science

University of British Columbia · Vancouver, BC · 2024

SKILLS

Languages & Frameworks: Java, JavaScript, TypeScript, Kotlin, Swift, Python, C#, C, C++, SQL, HTML, CSS, React, Node, Express, .NET, Jetpack Compose, Gradle, Blueprint, scikit-learn, CMake
Tools & Platforms: Git, Linux, Visual Studio, VS Code, Android Studio, Xcode, Unity, Unreal Engine, JIRA, Jenkins, Postman, Firebase, Docker, AWS CLI, IntelliJ, GitHub Actions
Systems, Cloud & Concepts: AWS (Lambda, Gateways, DynamoDB, S3), CI/CD, REST APIs, Data Structures & Algorithms, Multithreading, Client-Server Architecture, System & Database Design